SYSTEM DESIGN

Pacific Trade Intelligence Pipeline

How public shipping, trade and sanctions data are combined to describe semiconductor and critical-mineral imports through US Pacific ports, and to raise documented red flags for trade-based money laundering, sanctions evasion and forced labour.

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Version	0.3 (branch `pacific-trade-pipeline`)
Date	3 November 2026
Status	Built and tested on synthetic data; not yet run on live data
Language	Python 3.10+ (pandas, NumPy, PyArrow, requests, zstandard)
Source basis	Data-source recon of 19 October 2026 and the code as of this date

1. Purpose and scope

The pipeline answers three kinds of question about goods arriving by sea at US Pacific ports (Hawaii and the West Coast, with Guam as AIS-only context):

1. **What arrives, from where, and how.** Value and weight by port of unlading, HS6 product line, country of origin and month, split by vessel, container and air.
2. **How concentrated and exposed the flows are.** Supplier-country concentration per port, port concentration per product, global exporter concentration, port activity and congestion from ship positions, and a clearly labelled model of how much rides on individual ships.
3. **Where the patterns look wrong.** A library of red-flag checks drawn from published typologies (FATF, OFAC, US Department of Labor and others), each one documented with its sources and its innocent explanations.

Commodities in scope

Group	HS codes	Examples
Semiconductors and cable	8541, 8542, 8543, 8544	Diodes, transistors, LEDs and PV cells; integrated circuits; electrical machines n.e.s.; insulated wire and cable
Critical minerals	2609–2621, 2846	Tin, chromium, tungsten, molybdenum, titanium, niobium/tantalum ores; metal-bearing slags and residues; rare-earth compounds

Ports in scope

Census ports 3201 Honolulu (including Barbers Point, assumed), 3202 Hilo, 3203 Kahului, 3204 Nawiliwili–Port Allen, 3206 Kona, 2704 Los Angeles, 2709 Long Beach, 2809 San Francisco, 2811 Oakland, 3001 Seattle and 3002 Tacoma. Apra Harbor, Guam, appears in AIS only, because Guam is outside the Census customs territory.

The central limitation

No source in scope links a specific ship to a specific cargo. Census reports port × product × origin × month; AIS reports ship positions. The sources are joined at **port × month** and **origin country × HS6 × period** only. Anything said at ship level is an allocation model, and every red flag is a lead for review, never a finding.

Out of scope

- Bills of lading, manifests or any other shipment-level commercial data (CBP manifest data would close the ship-to-cargo gap, but it is not used).
- Profiling private individuals. The entities analysed are trade flows, trade lanes, countries and vessels; company names enter only through the analyst-supplied `screen` command.
- Satellite AIS. US coastal receivers only, so ocean legs are invisible.

2. Architecture

The system is a batch pipeline driven by a single command-line program. Each ingest command writes to a local data directory, every API response is cached on disk, and one `build` step turns whatever inputs are present into tables, a Markdown report and a self-contained HTML dashboard. Steps whose inputs are missing are skipped rather than failing the build.

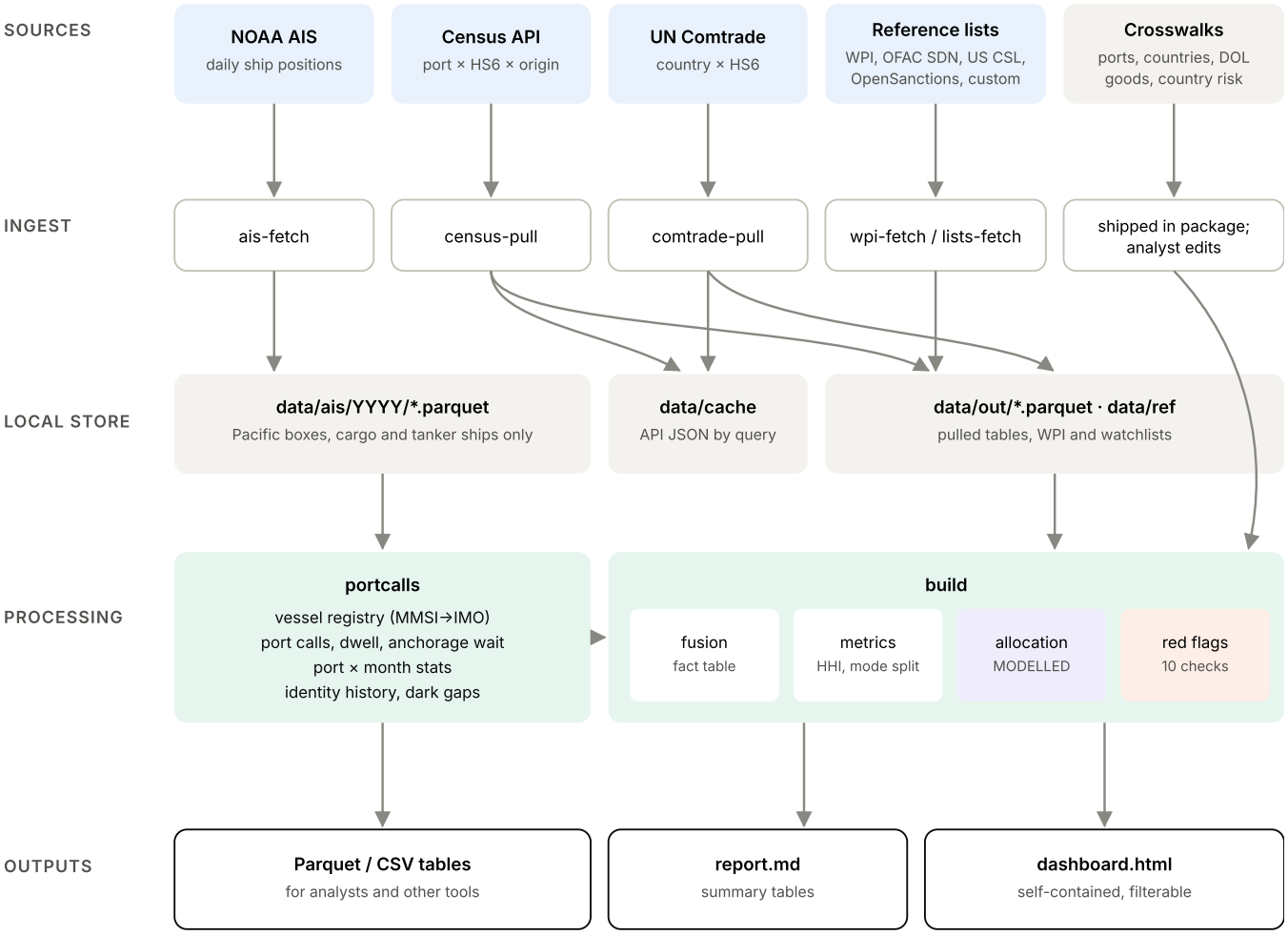


Figure 1. Data flows top to bottom. Every box in the ingest row is a CLI command; `build` is the only step that produces outputs.

Design principles

- **Re-runnable and cheap to repeat.** Processed AIS days are skipped; Census and Comtrade responses are cached under `data/cache` keyed by query without the API key, so re-runs cost no quota.
- **Degrade, don't fail.** Missing inputs skip the dependent tables and checks, and the run log says which and why.
- **Observed vs modelled is explicit.** Modelled outputs carry an `est_` prefix and `estimated=True`; the dashboard puts them under a "MODELLED, not observed" banner.

- **Every judgement is documented next to the code that makes it.** Scope constants live in `config.py`; each red-flag check carries its question, method, sources and innocent explanations.

3. Data sources

Source	Access	Auth and limits	Grain	Coverage and notes
NOAA Marine Cadastre AIS	Daily `.csv.zst` on the NOAA Azure blob (2025+); legacy daily `.zip` (≤2024)	None	Ship × position, about one per minute	2009 to 30 June 2026. US terrestrial receivers only (≈40–50 mi from shore); no satellite; Alaska removed after March 2021. ~200–300 MB compressed per day; schema changed in 2025.
Census International Trade API	`imports/porths` (port × HS6) and `imports/hs` (district × HS10)	Free key, required	Port of unlading × HS6 × country of origin × month	2010 onward. Vessel, containerised and air value and weight. Disclosure suppression at small ports. Annual revisions with April data.
UN Comtrade	`data/v1/get` (keyed) or keyless preview	Free key: 500 calls/day, 1 call/s	Reporter × partner × HS6 × month or year	US annual 2025, monthly to July 2026 at recon time. M49 country codes (Taiwan = 490). Sea/air split not reliable for the US.
World Port Index	NGA Pub 150 CSV	None	Port	Coordinates and UN/LOCODEs for the port geofences.
OFAC SDN list	CSV from OFAC	None	Listed party	Vessel IMO numbers parsed from the remarks field.
US Consolidated Screening List	CSV from trade.gov	None	Listed party	Combines Commerce, State and Treasury lists, including the BIS Entity List.
OpenSanctions (optional)	`targets.simple.csv`	Free for non-commercial use; licence for commercial use	Listed party	EU, UK, UN and many national lists in one schema.

The reference-list download URLs could not be reached from the build environment; they should be confirmed on the first live run.

Schema handling

AIS arrives in two layouts: the 2025+ snake_case schema (longitude before latitude, sentinel values already null) and the legacy CamelCase schema. Both are normalised to the 2025 layout; legacy sentinels (heading 511, speed 102.3, course 360, zero dimensions) become null, and the `IMO` prefix is stripped. Census responses are JSON arrays with a header row; predicate variables echoed back are de-duplicated and numeric fields coerced.

4. Data model and joins

Three sources, two join keys, and one gap that no key can close.

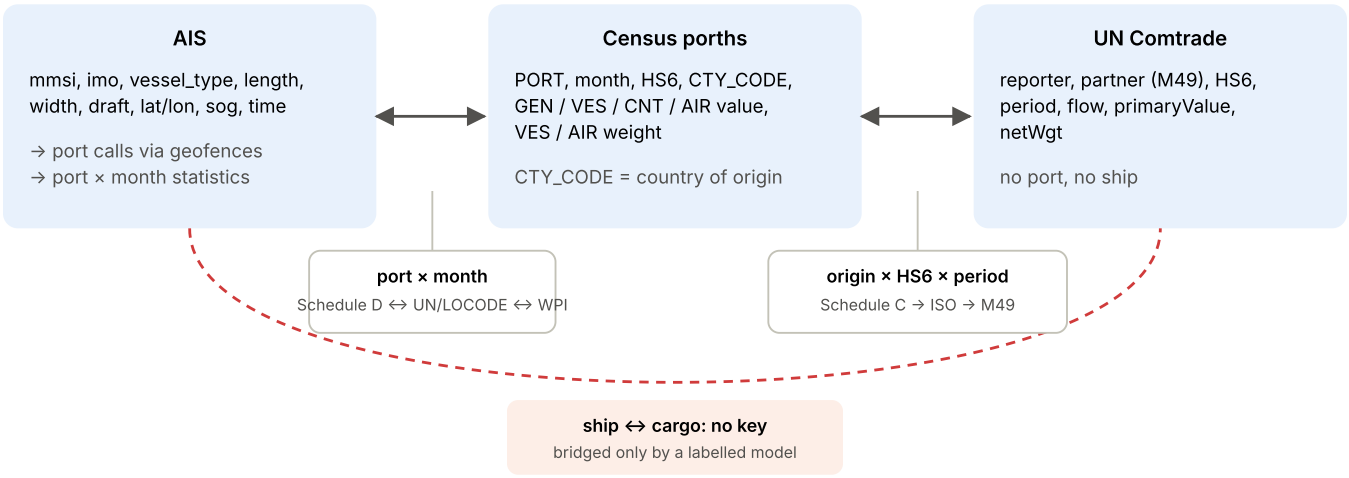


Figure 2. AIS joins Census on port × month through the port crosswalk and geofences. Census joins Comtrade on origin country × HS6 × period through the country crosswalk. Nothing joins a ship to its cargo.

Crosswalks

Table	Maps	Notes
`ports.csv`	Census Schedule D port → UN/LOCODE → WPI number → lat/lon → geofence radius → region	13 geofences. Several geofences can share a Census port (Honolulu and Barbers Point). Coordinates are overwritten from the World Port Index on `wpi-fetch`.
`countries.csv`	Census Schedule C `CTY_CODE` → ISO2/ISO3 → Comtrade M49	29 countries. Comtrade uses 842 for the US, 699 for India and 490 for Taiwan. `census-pull` prints any origin code that is not yet mapped.
`dol_goods_hs.csv`	DOL-listed good and country → HS prefixes → labour type	4 seed rows, unverified.
`country_risk.csv`	ISO3 → comprehensive sanctions, FATF status, Basel AML score, TIP tier, as-of date	Ships empty; the analyst fills it from current lists.

Grain of the fused fact table

`fact_port_month_hs6_origin`: one row per Census port × month × HS6 × origin country, carrying Census values, the AIS statistics for that port-month (`ais_*` columns), the origin's share of world exports for that HS6 and year, and the global exporter HHI for that HS6 and year. Census aggregate rows (the total and regional groupings) are excluded; only 4-digit country codes 1000–7999 are kept.

5. Pipeline stages and commands

All commands run as `python -m pacific_trade <command>` (or `pacific-trade` once installed). The order below is the normal run.

Command	Does	Reads	Writes
`wpi-fetch`	Downloads the World Port Index and prints the enriched port crosswalk	NGA	`data/ref/wpi.csv`
`ais-fetch`	Streams each national daily file through a zstd decoder (zips via a temp file), keeps the Pacific boxes and ship types 70–89, normalises the schema. Threaded; failed days are reported and retried on the next run.	NOAA	`data/ais/YYYY/YYYY-MM-DD.parquet`
`census-pull`	Port × HS6 × origin × month for the crosswalk ports and HS prefixes in scope. `--national` for all ports, `--hs10` for district HS10 detail, `--no-wildcard` if the `8541*` filter is rejected.	Census API	`census_porths.parquet` (+ national, HS10)
`comtrade-pull`	World exports by reporter for the year; the mirror pair (US-reported imports by partner, partners' reported exports to the US) annually and optionally monthly. HS codes are batched 20 per call.	Comtrade, Census output	`comtrade_world_exports`, `comtrade_us_imports`, `comtrade_partner_exports`
`lists-fetch`	Downloads OFAC SDN and the US CSL; `--opensanctions` adds OpenSanctions. Custom CSVs go in `lists/custom/`.	OFAC, trade.gov, OpenSanctions	`data/ref/lists/*.csv`
`portcalls`	Builds the vessel registry, detects port calls, aggregates to port × month, records identity history and dark gaps.	Filtered AIS days	`vessel_registry`, `port_calls`, `port_month_ais`, `vessel_identity`, `ais_gaps`

`build`	Fusion, metrics, allocation model, red flags; writes all tables, the report and the dashboard. `--method draft_drop` for the allocation sensitivity run, `--csv` for CSV copies.	Everything above that exists	16 tables, `report.md`, `dashboard.html`
`screen`	Fuzzy-matches a CSV of company or owner names against the downloaded lists.	Analyst CSV, lists	`screening_hits.csv`
`demo`	Runs the full build on synthetic data with planted anomalies, for previewing. All outputs carry a demo banner.	Nothing external	Same as `build`

6. Analytics

Vessel identity resolution

IMO is missing on about 65% of raw AIS rows. The registry takes, per MMSI, the most common non-null IMO, name, call sign and type over the whole period, and the median dimensions. The stable vessel key is `IMO#######` where known, otherwise `MMSI#########`. MMSIs broadcasting more than one IMO are flagged.

Port-call detection

Each point is assigned to the nearest port whose radius contains it. A visit is a run of points in one geofence, split when the ship leaves or goes silent for more than 6 h. A visit counts as a call when time at ≤ 0.5 kn reaches 2 h. Calls touching the edge of the date range are flagged as censored.

Anchorage wait

Slow time (≤ 0.5 kn) spent within 25 km of the port but outside the berth geofence, in the 72 h before the call's arrival, attached to the next call by the same ship at the same port. This is the only route-deviation signal coastal-only AIS supports.

Concentration metrics

Herfindahl–Hirschman Index on the 0–10,000 scale (over 2,500 = highly concentrated): supplier countries per port × HS6, ports per HS6, and world exporters per HS6 (Comtrade). Each comes with member count, top member and its share.

Mode split

Vessel, air and containerised shares of general import value per port × HS6, plus US dollars per kg by vessel. This answers the recon's second risk first: if chips mostly fly, the AIS side of the story applies mainly to cable and minerals.

Ship-level allocation (modelled)

MODELLED, not observed
For each port × month × HS6, Census vessel weight and value are spread across the dry-cargo calls (AIS types 70–79) arriving that month, in proportion to a size proxy. Tankers are excluded. Missing dimensions are filled with the port-month median.

```
capacity: weight_i ∝ length × width × arrival_draft draft_drop: weight_i ∝ length × width × max(arrival_draft − departure_draft, 0) (calls with no usable draft change: capacity × port-month median drop ratio)
```

Outputs: per-call edges (vessel × port × month × HS6 with estimated kg and USD), single-ship dependence per port × month × HS6 (vessel HHI, largest ship and its share), per-ship exposure ranked by estimated value, and the port-month-HS6 cells that had tonnage but no detected call (left unallocated rather than forced onto a ship).

7. Red-flag framework

A registry of documented checks, each emitting rows in one shared schema, then rolled up per entity. Adding a check means writing one decorated function and a test; it appears in the report, the dashboard and the check library automatically.



Figure 3. A check whose inputs are missing returns nothing and is logged as skipped; a check that raises an error is logged and does not stop the others.

The ten checks

ID	Check	Method (summary)	Scores
Trade-based money laundering			
`TBML-PRICE`	Unit-price outlier	Robust z of log($/kg) within HS6 (median, MAD floored at 0.1); flows ≥ 100 kg and ≥ $50k	0.40–0.95 by |z|, +0.1 for ≥ $1M
`TBML-MIRROR`	Mirror-trade gap	US-reported imports ÷ partner-reported exports to the US; flags outside 0.67–1.5 where either side ≥ $1M	0.35–0.95 by |log ratio|

`TBML-EXCEED`	Imports exceed origin's exports	Annual US imports by origin ÷ origin's exports to World; non-reporting countries skipped	0.60–0.95; 0.40 if none reported
`TBML-SURGE`	Origin-share surge	Recent 3-month share vs baseline: +15 pp and ≥ 3×	0.35–0.75, +0.2 if < 2% of world exports
Sanctions and export-control evasion			
`SANC-VESSEL`	Vessel on a list	IMO (check-digit validated) or MMSI exact match; name-only match for review	0.95; 0.50 name only
`SANC-IDENTITY`	Identity manipulation	MMSI with several IMOs; bad IMO check digit; flag (MID) change; three or more names; MMSI outside 201–775	0.40–0.60, highest applies
`SANC-DARK`	Dark period inside coverage	Gap ≥ 12 h, both ends within 30 km of a port, moved ≥ 5 km	0.35–0.75 by gap length
`SANC-COUNTRY`	High-risk jurisdiction	Origins under comprehensive sanctions, on FATF lists, or Basel AML score ≥ 6.5	0.90 / 0.80 / 0.45 / 0.40
Forced labour in supply chains			
`FL-DOL-GOODS`	DOL-listed good and origin	Origin × HS6 matched to the DOL TVPRA list via HS prefixes	forced 0.60, child 0.45, both 0.65; +0.1 for ≥ $1M
`FL-TIP`	TIP Tier 3 origin	Origin ranked Tier 3 in the State Department TIP report	0.35

Severity bands: high ≥ 0.75, medium ≥ 0.45, low below. Sources cited by the checks include the FATF TBML report (2006) and FATF/Egmont risk indicators (2021), Global Financial Integrity's mirror-trade method, FinCEN advisories, the OFAC/State/US Coast Guard Sanctions Advisory for the Maritime Industry (May 2020), the DOL List of Goods Produced by Child Labor or Forced Labor, the State Department TIP report, FATF jurisdiction lists and the Basel AML Index.

Entity roll-up

```
risk(entity) = 1 − Π over checks c of (1 − max score of c on that entity)
```

Several weak, independent signals outrank one weak signal but never reach certainty, and a check that fires every month counts once. Entities are trade flows (port ← origin · HS6), trade lanes (origin · HS6), origin countries and vessels.

8. Outputs and dashboard

Tables written by `build`

Table	Contents
`fact_port_month_hs6_origin`	Fused fact table (section 4)
`origin_hhi`, `port_hhi`, `global_exporter_hhi`	Concentration metrics
`mode_split`	Vessel / air / containerised shares and $/kg
`port_month_ais`	Calls, ships, dwell and anchorage wait per port-month
`census_vs_comtrade`	Census general imports vs Comtrade US-reported imports (needs the national pull)
`est_vessel_edges`, `est_vessel_spof`, `est_vessel_exposure`, `est_unallocated`	Modelled ship-level allocation
`red_flags`, `red_flag_entities`, `red_flag_runlog`, `typology_library`	Red-flag outputs

Dashboard

One self-contained HTML file: no external scripts, fonts or network calls, so it opens offline and can be emailed or hosted as a static page. The build embeds the tables the page needs as JSON (with `</` escaped), and the browser re-aggregates them as filters change.

Section	Shows
Filters	Month range, commodity group, HS6 line, port, red-flag area; every section below responds
Headline tiles	Vessel import value, all-mode value, share by vessel, vessel weight, supplier HHI with level, AIS cargo calls
Trade (Census)	Mode split, vessel value by port, countries of origin, monthly trend
Port activity (AIS)	Cargo calls per month; mean hours moored and waiting at anchor
Global supply (Comtrade)	World exporter HHI per HS6 against the 2,500 line
Red flags	Flags by check, highest-risk entities, all flags with evidence and caveats, check library with this build's run status
Vessel exposure (modelled)	Estimated value per ship and single-ship dependence, under a "MODELLED" banner

Every chart has a table view and keyboard-reachable tooltips; the page supports light and dark themes and phone-width screens. The categorical palette was checked with a colour-vision-deficiency validator. Because the file embeds its data, sharing it shares the numbers.

9. Reference data and governance

Item	State	Action before publishing
Port crosswalk	Seed coordinates; `verified=0`	Run `wpi-fetch`; fill UN/LOCODEs for Nawiliwili and Kona; confirm Barbers Point reports under 3201 from the first Census pull
Country crosswalk	29 countries; `verified=0`	Check against Census Schedule C "By ISO code"; add any codes `census-pull` reports as unmapped
DOL goods mapping	4 rows; `verified=0`	Check against the current DOL list and its HS mapping
Country risk	Empty by design	Fill from the latest FATF plenary, Basel AML Index, OFAC programmes and TIP report; record `as_of`
Watchlists	Downloaded on demand	Refresh before each run (`lists-fetch --refresh`); respect the OpenSanctions licence

The reference tables ship inside the package. `PACIFIC_TRADE_CROSSWALKS` points the pipeline at an edited copy without touching the installed files.

Licensing and terms

- NOAA AIS, Census, World Port Index, OFAC SDN and the US CSL are US government data with no access constraints.
- UN Comtrade free keys allow 500 calls a day; bulk downloads are a paid tier.
- OpenSanctions is free for non-commercial use only.
- Sources such as Equasis restrict automated collection and are deliberately not scraped.

10. Operations

Configuration

Setting	Purpose
`CENSUS_API_KEY`	Required for any Census pull
`COMTRADE_API_KEY`	Recommended; without it only the keyless preview works (one period and product per call, 500 records)
`PACIFIC_TRADE_DATA`	Data directory; defaults to `./data`
`PACIFIC_TRADE_CROSSWALKS`	Optional path to edited reference tables

Keys belong in environment variables or a secrets store, never in code or chat.

Network access needed

`api.census.gov`, `comtradeapi.un.org`, `noaaocm.blob.core.windows.net` (AIS 2025+), `coast.noaa.gov` (AIS ≤2024), `msi.nga.mil`, `sanctionslistservice.ofac.treas.gov`, `www.treasury.gov`, `data.trade.gov`, and optionally `data.opensanctions.org`.

Volume and run time

Item	Estimate
AIS download	~200–300 MB per day; ~20 GB per quarter; ~80 GB per year compressed
AIS filtering	~25 s per day (recon measurement); only the filtered Parquet is kept
Census, Comtrade, lists	Small; minutes per quarter; cached after the first pull
`build`	Seconds on a quarter (the synthetic demo builds in under 2 s)

Recommended first live run: all trade data and lists for one quarter, plus about one week of AIS, then scale AIS up once the outputs are checked.

Rate limits and resilience

- HTTP retries with exponential backoff on 429 and 5xx; downloads are written to a temporary file and renamed on completion.
- Comtrade calls are throttled to one per 1.1 s and capped at the daily budget per process.
- A failed AIS day does not stop the others; re-running fetches only what is missing.

11. Limitations and guardrails

Known limitations

Limitation	Effect and mitigation
No ship ↔ cargo link	Ship-level figures are modelled and labelled; unallocated tonnage is reported, not forced onto ships
Coastal-only AIS	No ocean legs, no ship-to-ship transfers at sea; Hawaii coverage is thin (16 cargo/tanker IMOs on the probe day)
Semiconductors may mostly fly	The mode split is shown first; AIS conclusions apply mainly to cable and minerals if the vessel share is low
Origin, not lading country	Census `CTY_CODE` is where goods were made; transshipment hubs are invisible in Census and must be inferred
Disclosure suppression	Small ports and narrow HS6 lines lose or move values; missing is not zero
HS6 breadth	Price checks can flag legitimate product-mix differences; HS10 detail (`--hs10`) is the follow-up

Crew-entered AIS static data	Identity flags can be typos; corroborate with a ship registry
Guam	AIS only; no Census trade data

Guardrails

> **A flag is a lead, not a finding**
> Output language is "matches typology X, with this evidence and these caveats". Credible leads go to bodies that can act: FinCEN, CBP e-Allegations, the DEA tip line, or the National Human Trafficking Hotline.

- **Organisations and vessels, not private individuals.** The pipeline builds no profiles of people.
- **No accusations in public.** Naming a company or person as a launderer from public-data anomalies risks defamation; corroborate and route through the proper channels.
- **Data collection within the rules.** APIs and bulk downloads only; no logins the analyst does not own; licence terms respected.
- **Demo output is marked.** Synthetic runs carry a banner on the dashboard and the report, because they show real country, port and vessel-style names next to random numbers.
- **Human trafficking** is covered through the forced-labour supply-chain angle only; victim-facing work (ads, social media) is left to specialist organisations.

12. Code layout, testing and roadmap

Package layout

Module	Responsibility
`config.py`	Scope constants (HS codes, boxes, thresholds), endpoints, paths, keys
`http.py`	Retries, on-disk JSON cache, atomic streamed downloads
`ais.py`	Download, normalise both schemas, stream-filter, vessel registry
`census.py`, `comtrade.py`	API clients with caching and throttling
`crosswalk.py`	Port and country crosswalks, WPI enrichment, geofence assignment
`portcalls.py`	Visit segmentation, call detection, anchorage wait, port-month stats
`fusion.py`, `metrics.py`, `allocation.py`	Fact table, concentration and mode metrics, modelled allocation
`redflags/`	`framework.py` (registry, schema, roll-up), `tbml.py`, `sanctions.py` (incl. list loaders, screening, identity and gap detection), `forced_labor.py`
`build.py`, `report.py`, `dashboard.py`	Assemble tables, render Markdown and HTML
`demo.py`, `cli.py`	Synthetic data with planted anomalies; command-line interface

Testing

40 offline tests (pytest), no network required: schema normalisation and sentinel handling, zstd streaming, port-call and anchorage logic, LA/Long Beach overlap, API parsing and caching (mocked), the missing-key redirect, allocation conservation, every red-flag check including IMO check digits and list loaders, the entity roll-up, error isolation between checks, dashboard payload escaping, and end-to-end `portcalls` → `build` and `demo` runs. A clean, non-editable install runs the demo end to end.

Roadmap

#	Item	Why
1	First live run (one quarter)	Validate against USA Trade Online by hand; learn the real false-positive rate of each check
2	Drug-precursor module	Precursor-chemical trade flows and mirror gaps; CBP seizure statistics; OFAC narcotics designations
3	Illegal fishing and maritime module	Global Fishing Watch satellite AIS (fixes the coastal-only blind spot), encounters and gaps, fisheries-body IUU vessel lists
4	Counterfeit and illicit-goods module	CBP IPR seizure statistics, OECD/EUIPO counterfeit-trade data
5	Ownership graph	GLEIF, OpenCorporates, UK Companies House beneficial owners, ICIJ Offshore Leaks; entity resolution across them
6	Validation set	Score the checks against publicly charged cases (DOJ, OFAC and BIS enforcement actions)
7	Standalone repository	Move the package out of `brand-intel` into its own repository for sharing